

A full L^2 solution of the matrix heat-composite wavelet inversion problem

Candidate solution packet for arXiv:0711.1424
Run `fa_banach_001`, lane 10, model GPT5.6

13 August 2026

Status. Candidate full solution to Problem A in the natural L^2 setting; likely valid, with cautious novelty. The result gives an exact spectral admissibility criterion, honest L^1 cone wavelets, and closed-form constants. It bypasses, rather than repairs, the paper's unresolved Fubini exchange in Problem C.

1 Source question

The source is I. A. Aliev, B. Rubin, S. Sezer, and S. B. Uyhan, “Composite Wavelet Transforms: Applications and Perspectives,” arXiv:0711.1424v1 (2007), published in *Contemporary Mathematics* 464 (2008), 1–25 [1]. Problem A appears on PDF page 21, in Section 8.2.

For $n \geq m$, let $\mathfrak{M}_{n,m} \simeq \mathbb{R}^{nm}$ and let $\mathcal{P}_m$ be the cone of positive-definite real symmetric $m \times m$ matrices. Put

$$d = \frac{m+1}{2}, \quad \mathrm{d}_* a = |a|^{-d} \mathrm{d}a,$$

where $|a| = \det a$. If H_t is the matrix heat semigroup, the source defines

$$\mathcal{H}_w f(x, a) = \int_{\mathcal{P}_m} H_{a^{1/2} \eta a^{1/2}} f(x) w(\eta) \mathrm{d}\eta.$$

It asks for a precise natural-space meaning of

$$f = c_{m,\alpha} \int_{\mathcal{P}_m} \mathcal{H}_w(I^\alpha f)(\cdot, a) |a|^{-\alpha/2} \mathrm{d}_* a \tag{A_\alpha}$$

and, at $\alpha = 0$, the reproducing formula

$$f = c_m \int_{\mathcal{P}_m} \mathcal{H}_w f(\cdot, a) \mathrm{d}_* a. \tag{A_0}$$

It also asks for wavelet functions and explicit constants.

2 Proof intuition

Heat flow diagonalizes under the Fourier transform. If $q = \xi' \xi$, the composite transform has multiplier

$$W(a^{1/2} q a^{1/2}), \quad W(s) = \int_{\mathcal{P}_m} e^{-\mathrm{tr}(s\eta)} w(\eta) \mathrm{d}\eta,$$

$$(8.26) \quad \mathcal{H}f(x, a) = \int_{\mathcal{P}_m} H_{a^{1/2}\eta a^{1/2}} f(x) w(\eta) d\eta, \quad x \in \mathfrak{M}_{n,m}, \quad a \in \mathcal{P}_m.$$

Taking into account an obvious similarity between (8.26) and the corresponding “rank-one” formula for $m = 1$, we call $\mathcal{H}f(x, a)$ the *composite wavelet transform of f associated to the heat semigroup H_t* . Here w is a symmetric integrable function on $\mathcal{P}_m$ (that will be endowed later with some cancelation properties) and a is a $\mathcal{P}_m$ -valued scaling parameter. One can replace w by a more general *wavelet measure*, as we did in the previous sections, but here we want to minimize technicalities.

Owing to (8.10), it is natural to expect that the inverse of I^α has the same form (8.24) with α formally replaced by $-\alpha$, and the case $\alpha = 0$ gives a variant of Calderón’s reproducing formula.

Thus, we encounter the following open problem:

Problem A. *Give precise meaning to the inversion formula*

$$(8.27) \quad f(x) = c_{m,\alpha} \int_{\mathcal{P}_m} \frac{\mathcal{H}\varphi(x, a)}{|a|^{\alpha/2}} d_* a, \quad \varphi = I^\alpha f,$$

and the reproducing formula

$$(8.28) \quad f(x) = c_m \int_{\mathcal{P}_m} \mathcal{H}f(x, a) d_* a,$$

say, for $f \in L_p$ or any other “natural” function space. Give examples of wavelet functions w for which (8.27) and (8.28) hold. Find explicit formulas for the normalizing coefficients $c_{m,\alpha}$ and c_m , depending on w .

Solution of this problem would give a series of pointwise inversion formulas

Figure 1: Problem A and formulas (8.27)–(8.28), source PDF page 21.

the cone Laplace transform of w . Spectral invariance replaces $a^{1/2}qa^{1/2}$ by $q^{1/2}aq^{1/2}$. The congruence substitution $r = q^{1/2}aq^{1/2}$ then cancels exactly the Riesz multiplier $|q|^{-\alpha/2}$. Reconstruction is reduced to one scalar constant,

$$C_\alpha = \int_{\mathcal{P}_m} |r|^{-\alpha/2} W(r) d_* r.$$

When $W \geq 0$, balanced cone truncations make the resulting Fourier multipliers increase from 0 to C_α , so Plancherel and dominated convergence finish the proof without the Fubini step that blocked the source.

To obtain actual wavelets, differentiate a matrix-gamma density with the Cayley determinant operator. Its Laplace transform becomes $|s|^N |I_m + s|^{-\beta}$. The determinant factor provides cancellation on *every* rank-deficient boundary stratum; the remaining constant is a matrix beta integral.

3 The exact Hilbert-space criterion

Call a function on $\mathcal{P}_m$ *spectral* if it is invariant under orthogonal conjugation. This is the operationally well-defined version of the source’s “symmetric” condition. For $R > 1$, set

$$\Omega_R = \{a \in \mathcal{P}_m : R^{-1}I_m < a < RI_m\}.$$

For $\alpha \geq 0$, define the dense natural domain

$$\mathcal{D}_\alpha = \left\{ f \in L^2(\mathfrak{M}_{n,m}) : |\xi' \xi|^{-\alpha/2} \widehat{f}(\xi) \in L^2 \right\}.$$

For $f \in \mathcal{D}_\alpha$, $I^\alpha f \in L^2$ is defined by the displayed Fourier multiplier, in agreement with the source’s convention.

Theorem 1 (Spectral characterization). *Let $w \in L^1(\mathcal{P}_m)$ be spectral, and let*

$$W(s) = \int_{\mathcal{P}_m} e^{-\text{tr}(s\eta)} w(\eta) d\eta.$$

Assume $W(s) \geq 0$ on $\mathcal{P}_m$. Fix $\alpha \geq 0$ and put

$$C_\alpha(w) = \int_{\mathcal{P}_m} |r|^{-\alpha/2} W(r) \, d_* r. \quad (1)$$

If $0 < C_\alpha(w) < \infty$, then for every $f \in \mathcal{D}_\alpha$,

$$f = \frac{1}{C_\alpha(w)} \lim_{R \rightarrow \infty} \int_{\Omega_R} \mathcal{H}_w(I^\alpha f)(\cdot, a) |a|^{-\alpha/2} \, d_* a. \quad (2)$$

For $\alpha = 0$ this holds for every $f \in L^2(\mathfrak{M}_{n,m})$ and is the requested reproducing formula. Within the class $W \geq 0$, condition $0 < C_\alpha(w) < \infty$ is also necessary for (2), with balanced truncations, to reconstruct every function in the domain with a finite nonzero scalar normalization.

Proof. For fixed a , the defining integral for $\mathcal{H}_w$ is a Bochner integral in L^2 , because the heat operators are contractions and $w \in L^1$. The finite a -integral over Ω_R is therefore also a genuine L^2 Bochner integral. The heat multiplier from [1, formula (8.15)] gives, with $q = \xi' \xi$,

$$\widehat{\mathcal{H}_w g}(\xi, a) = W(a^{1/2} q a^{1/2}) \widehat{g}(\xi). \quad (3)$$

Orthogonal invariance of w implies the same invariance for W . When $q > 0$, the symmetric matrices $a^{1/2} q a^{1/2}$ and $q^{1/2} a q^{1/2}$ have the same eigenvalues, and hence

$$W(a^{1/2} q a^{1/2}) = W(q^{1/2} a q^{1/2}). \quad (4)$$

The set of $\xi \in \mathfrak{M}_{n,m}$ for which q is singular has Lebesgue measure zero.

Let T_R denote the truncated operator in (2), before division by C_α . Since $\widehat{I^\alpha f} = |q|^{-\alpha/2} \widehat{f}$, equations (3)–(4) and the invariant change of variables $r = q^{1/2} a q^{1/2}$ yield

$$\widehat{T_R I^\alpha f}(\xi) = M_R(q) \widehat{f}(\xi), \quad M_R(q) = \int_{q^{1/2} \Omega_R q^{1/2}} |r|^{-\alpha/2} W(r) \, d_* r. \quad (5)$$

Indeed, $d_* a = d_* r$ and $|a|^{-\alpha/2} = |q|^{\alpha/2} |r|^{-\alpha/2}$, which cancels the Riesz factor.

For every full-rank q , the sets $q^{1/2} \Omega_R q^{1/2}$ are nested and exhaust $\mathcal{P}_m$. Positivity gives

$$0 \leq M_R(q) \uparrow C_\alpha(w).$$

Thus $|M_R(q)| \leq C_\alpha(w)$ and Plancherel plus dominated convergence proves (2).

For necessity, the same pointwise monotone limit is (1). If it is zero, all limits vanish. If it is infinite, then $M_R(q) \rightarrow \infty$ at every full-rank frequency; choosing a nonzero Fourier transform supported in a compact full-rank set and applying Fatou shows that no finite normalization can give L^2 convergence. This proves the characterization. $\square$

Remark 2 (All- L^2 operator extension). Formula (5) defines $T_R I^\alpha$ as a uniformly bounded operator on all of L^2 , even when $I^\alpha f$ is not itself an L^2 function. Hence (2) extends canonically to every $f \in L^2$ in the composed-operator sense. The dense domain $\mathcal{D}_\alpha$ is used in the theorem so that every term in the source's displayed formula is an ordinary L^2 object.

4 Explicit integrable wavelets and constants

Recall the matrix gamma and beta functions

$$\Gamma_m(z) = \pi^{m(m-1)/4} \prod_{j=0}^{m-1} \Gamma(z - j/2), \quad B_m(u, v) = \frac{\Gamma_m(u) \Gamma_m(v)}{\Gamma_m(u+v)}.$$

They converge in their integral forms when the real parts of all parameters exceed $d - 1 = (m - 1)/2$ [1, formulas (8.4)–(8.5)].

On the vector space of symmetric matrices, use the normalized derivative matrix

$$\partial_{ii} = \frac{\partial}{\partial \eta_{ii}}, \quad \partial_{ij} = \frac{1}{2} \frac{\partial}{\partial \eta_{ij}} \quad (i < j), \quad \mathcal{D} = \det(\partial_{ij}).$$

For real $\beta > d - 1$, let

$$G_\beta(\eta) = \frac{\mathbf{1}_{\mathcal{P}_m}(\eta)}{\Gamma_m(\beta)} e^{-\text{tr } \eta} |\eta|^{\beta-d}. \quad (6)$$

Lemma 3 (Matrix-gamma Rodrigues wavelet). *Let $N \geq 1$ be an integer and suppose*

$$\beta - N > d - 1. \quad (7)$$

Then the distributional derivative

$$w_{N,\beta} = \mathcal{D}^N G_\beta \quad (8)$$

is represented by an ordinary spectral function in $L^1(\mathcal{P}_m)$. Its cone Laplace transform and cancellation are

$$W_{N,\beta}(s) = |s|^N |I_m + s|^{-\beta} \geq 0, \quad \int_{\mathcal{P}_m} w_{N,\beta}(\eta) d\eta = 0. \quad (9)$$

Proof. The standard Cayley/Riesz-distribution identity on a symmetric cone [3, Chapter VII] gives the matrix Rodrigues formula

$$\mathcal{D}^N \left(e^{-\text{tr } \eta} |\eta|^{\beta-d} \right) = e^{-\text{tr } \eta} |\eta|^{\beta-d-N} P_{N,\beta}(\eta), \quad (10)$$

on $\mathcal{P}_m$, where $P_{N,\beta}$ is a conjugation-invariant polynomial (a matrix Laguerre polynomial). In the range (7), the right side is integrable: the exponential handles infinity and the matrix-gamma criterion handles the boundary. The same criterion, applied through orders $0, \dots, N$, removes all boundary distributions from the zero extension in (6). Thus (8) is the ordinary L^1 function in (10), divided by $\Gamma_m(\beta)$.

The matrix-gamma integral gives

$$\int_{\mathcal{P}_m} e^{-\text{tr}(s\eta)} G_\beta(\eta) d\eta = |I_m + s|^{-\beta}. \quad (11)$$

Applying the constant-coefficient distribution $\mathcal{D}^N$ and integrating by parts multiplies its Laplace transform by $|s|^N$ (the two signs $(-1)^{mN}$ cancel). This proves (9). Finally, setting $s = 0$ in (9) gives the cancellation integral. $\square$

For orientation, when $m = 1$ formula (8) is simply

$$w_{N,\beta}(t) = \frac{1}{\Gamma(\beta)} \frac{d^N}{dt^N} (e^{-t} t^{\beta-1}).$$

When $m = 2$, $N = 1$, and $\lambda = \beta - 3/2$, it becomes

$$w_{1,\beta}(\eta) = \frac{e^{-\text{tr } \eta} |\eta|^{\beta-5/2}}{\Gamma_2(\beta)} \left(|\eta| - \lambda \text{tr } \eta + \lambda(\lambda + \tfrac{1}{2}) \right).$$

Theorem 4 (Closed-form answer to Problem A). *Let $n \geq m$, $\alpha \geq 0$, and choose an integer N and a real β such that*

$$N > \frac{\alpha + m - 1}{2}, \quad \beta > N + \frac{m - 1}{2}. \quad (12)$$

Then the integrable wavelet $w_{N,\beta}$ in (8) satisfies the inversion formula (2), with

$$c_{m,\alpha} = \frac{\Gamma_m(\beta)}{\Gamma_m(N - \alpha/2)\Gamma_m(\beta - N + \alpha/2)}. \quad (13)$$

At $\alpha = 0$,

$$c_m = \frac{\Gamma_m(\beta)}{\Gamma_m(N)\Gamma_m(\beta - N)}. \quad (14)$$

In particular, $N = m$ and $\beta = 2m$ give one reproducing wavelet for every $m \geq 1$.

Proof. By (9), the constant in Theorem 1 is

$$\begin{aligned} C_\alpha(w_{N,\beta}) &= \int_{\mathcal{P}_m} |r|^{N-\alpha/2-d} |I_m + r|^{-\beta} dr \\ &= B_m \left(N - \frac{\alpha}{2}, \beta - N + \frac{\alpha}{2} \right). \end{aligned} \quad (15)$$

The second equality is the beta-prime form of the matrix beta integral, obtained from the source's beta integral by $r = t(I_m - t)^{-1}$. Conditions (12) put both beta parameters above $d - 1$ and imply the L^1 condition (7). Equations (13)–(14) are the reciprocal of (15), and Theorem 1 applies. $\square$

5 What happens to Problems B and C?

Problem B asks for examples satisfying the source's particular Definition 8.7, formulated through a Gårding–Gindikin function $\Lambda_{\alpha/2}$. The packet does not prove that $w_{N,\beta}$ belongs to that narrower class. Instead, Theorem 1 supplies a different exact spectral characterization that is sufficient and necessary for the balanced L^2 formula when $W \geq 0$.

Problem C records an unjustified interchange in the proposed Gårding–Gindikin calculation. That interchange is absent here. All finite-truncation integrals are Bochner integrals in L^2 , and the only limit step is dominated convergence of bounded nonnegative Fourier multipliers. Thus Problem C ceases to obstruct Problem A for this family, but the packet does not classify all additional hypotheses that would repair the source's original pointwise Fubini argument.

$$k(s) = \frac{\Gamma_m(d)}{|s|^d \Gamma_m(\alpha/2 + d)} \int_0^s w(b) |s - b|^{\alpha/2} ds = \frac{\Gamma_m(d)}{|s|^d} I_+^{\alpha/2+d} w(s) = \Lambda_{\alpha/2}(s).$$

Problem C. Although all calculations above go through smoothly, interchange of the order of integration remains unjustified. We do not know how to justify it and what additional requirements on the wavelet w should be imposed (if any). One of the obstacles is that $\int_0^\infty \neq \int_0^s + \int_s^\infty$, when we integrate over the higher-rank cone.

References

- [Al] I.A.Aliev, *On the Bessel type potentials and associated function spaces*, Preprint, 2007.
- [AB1] I.A.Aliev and S. Bayrakci, *On inversion of B-elliptic potentials by the method of Balakrishnan–Rubin*, *Fract. Calc. Appl. Anal.* **1** (1998) 265–284.

Figure 2: The Fubini obstruction stated as Problem C, source PDF page 23.

6 Verification, limitations, and novelty

The proof was audited along eight focused routes: exact source scope; comparison with the direct-wavelet predecessor [2]; the general spectral reduction; necessity of the weighted criterion; the boundary-free

Cayley derivative; the matrix-beta constant; the L^2 domain and truncation; and the Problem B/C scope distinction. A symbolic checker verifies the first $m = 2$ Rodrigues identity and scalar numerical specializations; no computation is proof-critical.

The most delicate review point is Lemma 3, specifically the absence of boundary distributions under (7). This is the standard safe range of the determinant Riesz distribution and is stronger than needed for some individual parameters. The proof deliberately does not assert pointwise convergence or an L^p theorem for $p \neq 2$.

Cheap run indexes, exact title/arXiv/problem searches, cone-heat-wavelet keyword searches, and the full source of arXiv:math/0409100 were checked. No later exact answer was located through 2026-08-13. The predecessor proves a related direct multiscaled-wavelet theorem but assumes, for Riesz inversion, a Fourier profile supported away from the rank-deficient boundary. A nonzero cone Laplace profile cannot satisfy that hypothesis. Novelty remains cautious because the present multiplier argument is philosophically close to that predecessor and the Rodrigues family may exist in matrix-Laguerre language.

Human-review recommendation. High-priority expert review. Verify the Cayley/Riesz safe-range statement, then check the scope judgment that an L^2 solution is a full answer to the source’s phrase “ L_p or any other natural function space.”

References

- [1] I. A. Aliev, B. Rubin, S. Sezer, and S. B. Uyhan, *Composite Wavelet Transforms: Applications and Perspectives*, arXiv:0711.1424v1 (2007); Contemp. Math. 464 (2008), 1–25.
- [2] G. Ólafsson, E. Ournycheva, and B. Rubin, *Multiscaled wavelet transforms, ridgelet transforms, and Radon transforms on the space of matrices*, arXiv:math/0409100; Appl. Comput. Harmon. Anal. 21 (2006), 182–203.
- [3] J. Faraut and A. Korányi, *Analysis on Symmetric Cones*, Oxford Mathematical Monographs, Clarendon Press, 1994.